**25901280** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

## Fourth SEMESTER

## Discipline Specific Core Course (DSC) - MG4DSCFSC202 - FORENSIC BIOLOGY

(2024 ADMISSION ONWARDS)

**Duration: 1 hour 30 minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Mention any four divisions of forensic biology. [K] / [CO1]
2. Explain the forensic importance of amylase detection in saliva. [K] / [CO3]
3. What colour change indicates a positive Phenolphthalein test? [K] / [CO5]
4. Describe acid digestion method for diatom extraction. [S] / [CO4]
5. Describe microscopic examination of fibre cross-sections. [An] / [CO2]
6. Analyze significance of medullary index in species identification. [An] / [CO2]
7. How can stringing method help determine the point of origin? [S] / [CO5]
8. Why is the Benzidine test no longer preferred? [K] / [CO5]
9. Explain biochemical composition of hair. [U] / [CO2]
10. What is the confirmatory test for seminal stains? [U] / [CO3]

11. Differentiate between Point of Convergence and Point of Origin. [U] / [CO5]

12. State what is meant by a diatom [K] / [CO4]

**[2x10 = 20]**

## Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13. Define collection of biological evidence. [U] / [CO3]

14. Explain the principle of Acid Phosphatase test. [U] / [CO3]

15. Demonstrate how to extract diatoms from lung tissue for forensic examination. [A] / [CO4]

16. Describe laboratory comparison of questioned and known fibres. [S] / [CO2]

17. Analyze why diatoms are more reliable than stomach content for confirming drowning. [An] / [CO4]

18. Explain the biological and physical properties of blood that influence stain formation. [U] / [CO5]

19. Explain the biochemical basis for identification of blood at a crime scene. [U] / [CO5]

**[4x5 = 20]**

## Part C

Essay questions

Answer any **1** questions

Each question carries **10** marks

20. Describe the forensic significance, collection and preservation of other body fluids such as saliva, sweat and faecal matter in criminal investigations. [A] / [CO2]

21. Describe detailed laboratory comparison of human and animal hair. [S] / [CO2]

**[10x1 = 10]**